
MA 524: Probability and Statistics

Course Code	MA 524
Course Name	Probability and Statistics
L-T-P-C	3-1-0-4
Prerequisites	NA
Intended for	M.Sc./B. Tech./M.S./Ph.D.
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Probability and random variable**
 σ field; measurable space; construction of measure probability and properties; definitions, scope and examples of probability; sample spaces and events; axiomatic definition of probability; joint and conditional probabilities; independence, total probability; Bayes' rule and applications. [8 Lectures]
Definition of random variables, continuous and discrete random variables; cumulative distribution function (cdf) for discrete and continuous random variables; probability mass function (pmf); probability density functions (pdf) and properties; expectation: mean, variance and moments of a random variables. [5 Lectures]
- **Distribution Functions** [7 Lectures]
Some special distributions: uniform, exponential, Chi-square, Gaussian, binomial, and poisson distributions; Law of large numbers; Central limit theorem and its significance.
- **Statistics** [6 Lectures]
Scatter diagram; graphical residual analysis, Q-Q plot to test for normality of residuals, autocorrelation and autocovariance functions; stationarity and non stationarity ; correlation and covariance
- **Module** [12 Lectures]
Sampling distributions; point and interval estimation, testing of hypothesis, Goodness of fit and contingency tables, linear regression, ANOVA.
- **Module** [4 Lectures]
Introduction to Stochastic process; white noise; random walk and Brownian motion.

Text Books

1. Sheldon M. Ross, **Introduction to Probability and Statistics for Engineers and Scientists**, Academic Press, 2009.

References

1. D. C. Montgomery and G.C. Runger, **Applied Statistics and Probability for Engineers**, 5th Edition, John Wiley & Sons, 2009.
2. Robert H. Shumway and David S. Stoffer, **Time Series Analysis and Its Applications with R Examples**, 3rd Edition, Springer Texts in Statistics, 2006.